

RADAR-IoT: An open-source, interoperable and extensible IoT gateway framework for healthcare and beyond

Yatharth Ranjan
Institute of Psychiatry Psychology
and Neuroscience, King's College
London
London, UK
yatharth.ranjan@kcl.ac.uk

Heet Sankesara
Institute of Psychiatry Psychology
and Neuroscience, King's College
London
London, UK
heet.sankesara@kcl.ac.uk

Pauline Conde
Institute of Psychiatry Psychology
and Neuroscience, King's College
London
London, UK
pauline.conde@kcl.ac.uk

Jiangeng Chang
Saw Swee Hock School of Public
Health, National University of
Singapore
Singapore, Singapore
e0950658@u.nus.edu

Zulqarnain Rashid
Institute of Psychiatry Psychology
and Neuroscience, King's College
London
London, UK
zulqarnain.rashid@kcl.ac.uk

Richard Dobson
Institute of Psychiatry Psychology
and Neuroscience, King's College
London
London, UK
richard.dobson@kcl.ac.uk

Amos A Folarin
Institute of Psychiatry Psychology
and Neuroscience, King's College
London
London, UK
amos.folarin@kcl.ac.uk

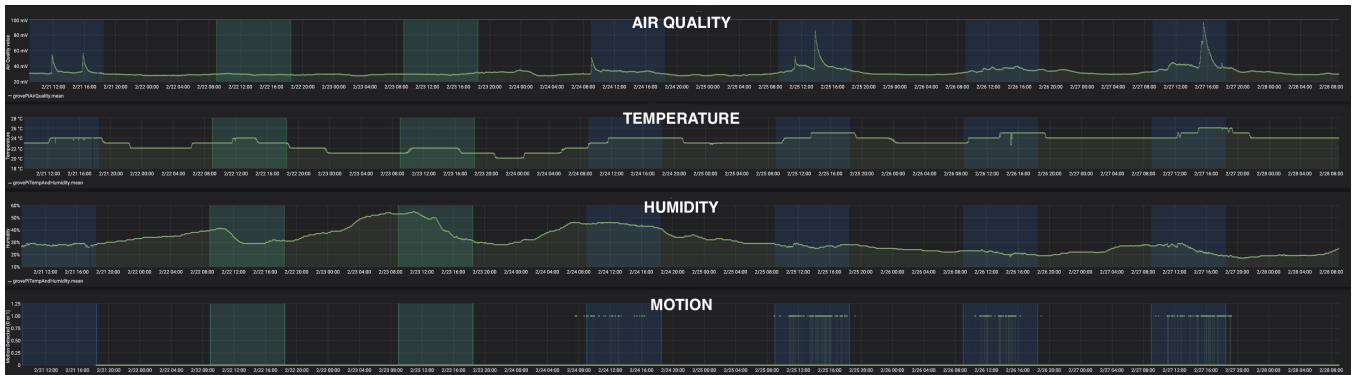


Figure 1: Real-time dashboard for data collected from RADAR-IoT.

ABSTRACT

RADAR-IoT is an IoT gateway framework allowing myriad of sensors to inter-connect, collect data and do interesting things like on-device AI/ML, dashboarding, alerting and combining sensor information with other data sources of a cloud based platform (like wearable devices in an mHealth data platform) and providing real-time actionable intelligence. The amount of available sensors is

limitless allowing for possibilities previously not possible, empowering the field of health informatics.

KEYWORDS

internet of things, sensors, health, mHealth, mobile health, real time, iot, iomt, iot gateway

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

Conference acronym 'XX, June 03–05, 2018, Woodstock, NY

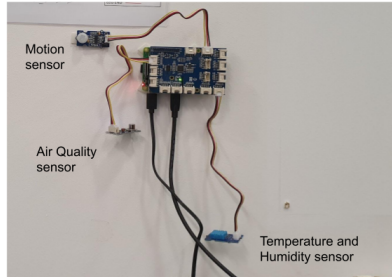
© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9423-9/22/09.

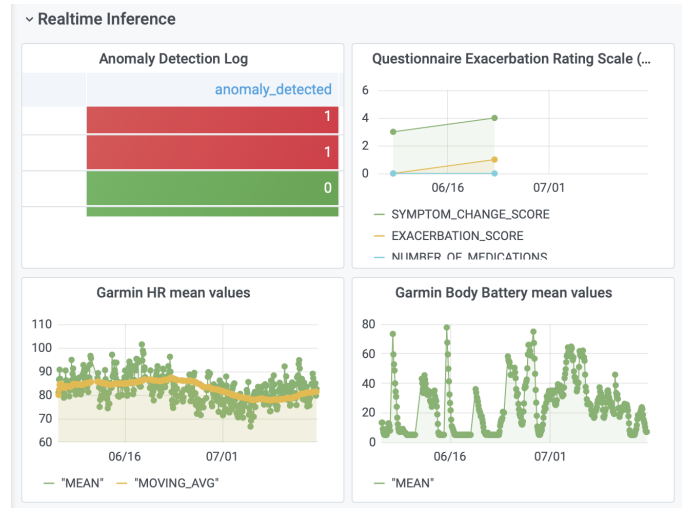
<https://doi.org/10.1145/3544793.3560318>

ACM Reference Format:

Yatharth Ranjan, Heet Sankesara, Pauline Conde, Jiangeng Chang, Zulqarnain Rashid, Richard Dobson, and Amos A Folarin. 2022. RADAR-IoT: An open-source, interoperable and extensible IoT gateway framework for healthcare and beyond. In *Woodstock '18: ACM Symposium on Neural Gaze Detection, June 03–05, 2018, Woodstock, NY*. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3544793.3560318>



(a) RADAR-IoT gateway with sensors



(b) Coherent view of data on RADAR-Base platform

Figure 2: RADAR-IoT setup and dashboard

1 INTRODUCTION

The Internet of Things (IoT) describes the network of physical objects—“things”—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet. An IoT gateway device inter-connects all the IoT devices at a location (called at-edge) to the cloud based platforms and IoT devices in other locations. It’s functionality can also include rudimentary processing of the data and security and privacy ensuring access.

There are various limitations with current IoT gateway frameworks. These include being closed source, limited number or type of sensors and I/O protocols, restricted interoperability and scalability, no demonstrable support of cloud platforms and limited proof of real world usage.

The proposed IoT gateway framework, RADAR-IoT, differentiates itself from related work by being device, sensor and programming language agnostic, supporting all types of common IoT input-output protocols, being open-source, modular and easily extensible, providing support for multiple data sinks (like mHealth platform, on-device AI and ML, dashboard and more), interoperability, providing industry-leading security and medical level privacy, and providing integration to a well established open-source mHealth cloud platform for data collection, aggregation, transformation and heavyweight analytics with different types of data sources like wearables, IoT sensors, mobile apps, eCRFs.

2 RADAR-BASE PLATFORM

RADAR-base (Remote Assessment of Disease And Relapses) [1] is an open source platform to leverage data from wearables and mobile technologies. The main focus of RADAR-base is the seamless integration of data streams from various wearables to collect sensor data in real time and store, manage and share the collected data with researchers for retrospective analysis.

RADAR-base provides both passive and active data collection via two applications. Passive data collection includes real time monitoring of movement, location, audio, calls and texts, and app usage. Active data collection includes the use of questionnaires that might ask patients about their mood, medication intake, or the severity of symptoms. It also supports real time streaming ETL and analysis of the data and taking actions (such as interventions) based on results from the real time analysis or inference as shown in Fig 2b.

Presently, RADAR-base is being used in an array of research studies of all types and sizes. It currently focuses on personal sensing; these devices are typically battery powered and carried on the user. Wearable devices available for integration are limited by the vendor availability of SDKs and REST-APIs, however, for static IoT sensors there is very large range of sensor modalities and providers, a significant improvement to the RADAR-base platform would enable the use of these sensors opening up a wide array of use cases within the health and other domains. Moreover, these IoT sensors can be installed in static settings, meaning they can be connected to power continuously, removing the limitations that mobile devices face with respect to battery and resources. These are particularly useful in cases where a holistic view of a patient’s health is required including their environment and surroundings. RADAR-IoT framework aims to add this capability to the RADAR-Base mHealth platform while also being flexible enough to be used with other platforms and providers for different use-cases.

3 RADAR-IOT

RADAR-IoT is a flexible and extensible Internet of Things edge gateway framework for plugging in a wide variety of sensors and provision of flexible data processing of those inputs and with support for uploading IoT sensor data to RADAR-base platform. RADAR-IoT also supports uploading to InfluxDb to provide real time visualisation through Grafana and can be easily extended to include

more backend/cloud destinations. All the components are based on abstract sub-components streamlining addition of more data processing variants and sensors.

4 DEMONSTRATION

The demonstration of the RADAR-IoT framework for RADAR-Base will include a gateway device (a Raspberry Pi 4B) with a Grove Pi Plus hat. Four sensors will be attached to the hat including Air Quality, Temperature and Humidity, Light and Motion sensors as shown in Fig 2a. All of this apparatus will be neatly packaged in a case with an attached touch screen that will display the data from the sensors.

The data from these sensors will be streamed in real-time to an InfluxDb and Grafana dashboard (shown in Fig 1) and also to the RADAR-Base platform. Alongside this, data (heart rate, SpO2 and more) from a wearable device will also be streamed in real-time

to the RADAR-Base mHealth platform. The data on the RADAR-Base platform will be displayed through a dashboard on which a coherent view of information from other sources such as a wearable device will be combined with IoT sensor data to provide unique insights shown in Fig 2b. The dashboards and information will be displayed on a large screen at the venue.

The demonstration will be interactive where the attendees can take actions that will trigger change in sensor values that can be seen in real-time on the dashboards with a corresponding change in their wearable data such as heart rate. This can also be combined with a real-time inference module and send notifications/alerts on detection of any anomalies in sensor values.

REFERENCES

- [1] Yatharth Ranjan, Zulqarnain Rashid, Callum Stewart, Pauline Conde, Mark Begale, Denny Verbeeck, Sebastian Boettcher, Richard Dobson, Amos Folarin, RADAR-CNS Consortium, et al. 2019. RADAR-base: open source mobile health platform for collecting, monitoring, and analyzing data using sensors, wearables, and mobile devices. *JMIR mHealth and uHealth* 7, 8 (2019), e11734.